

Inside the Universe

Where Do the Zeta Zeros Emerge From?

The 0/0 Structure of Cosmic Growth

We show that the Riemann zeta zeros are the missing fourth component of the universe's interior (after matter, radiation, and dark energy).

They emerge from the 0/0 singularity of the Euler product at $\text{Re}(s)=1/2$.

The universe grows by superimposing the zeta zero harmonics.

1. The Logarithmic 0/0 Structure

Two fundamental conventions define the 0/0 framework:

$\log_1(1) = 0$ The tautology: $1^0 = 1$
 The boundary of knowledge.
 The universe ENDS here.

$\log_0(0) = x$ The singularity: $0^x = 0$ for all $x > 0$
 The origin of being.
 The universe BEGINS here.

KEY: $\log_0(0)$ is undefined in standard mathematics.

In the 0/0 framework, it is the REMOVABLE SINGULARITY at the origin of existence. The removable value is the universe itself.

Zeta Function at Special Points

At $s = 1$ (the pole):

$\zeta(s) = 1/(s-1) + \gamma + O(s-1)$
 $\Rightarrow 0/0$ structure: numerator = 1, denominator = $s-1 \rightarrow 0$
 $\Rightarrow$ Removable value: γ (Euler-Mascheroni = 0.5772...)

At $s = 0$ (the functional equation):

$\zeta(s) = 2^s \cdot \pi^{s-1} \cdot \sin(\pi s/2) \cdot \Gamma(1-s) \cdot \zeta(1-s)$
 At $s=0$: $\sin(0) = 0$, $\zeta(1) = \text{infinity}$
 $\Rightarrow 0 \cdot \text{infinity} = -1/2$ (removable!)

At $s = 1/2 + i\gamma_n$ (the zeros):

$\zeta(s) = 0$
 $\Rightarrow$ The functional equation creates a $0/0$:
 both sides vanish simultaneously
 $\Rightarrow$ The removable value is 0 (the zero itself)

2. What is Inside the Universe

The Poincare sphere maps all of spacetime onto a finite ball.

The INTERIOR contains four components:

1. MATTER: $\rho_m = \rho_{m0} / a^3$
 Growth: $a(t) \sim t^{2/3}$
 $0/0$: $a=0$ at $t=0$ (Big Bang, removable)
2. RADIATION: $\rho_r = \rho_{r0} / a^4$
 Growth: $a(t) \sim t^{1/2}$
 $0/0$: $a=0$ at $t=0$ (Big Bang, removable)
3. DARK ENERGY: $\rho_L = \Lambda/(8\pi G) = \text{const}$
 Growth: $a(t) = \exp(H \cdot t)$
 $0/0$: Λ is the removable value of vacuum energy
4. THE PRIME SPECTRUM: zeta zeros as standing waves
 Growth: each zero contributes an oscillation to $\psi(x)$
 $0/0$: the zeros emerge from the Euler product singularity

THE ZETA ZEROS ARE THE FOURTH COMPONENT.

Matter, radiation, dark energy, and the PRIME SPECTRUM.

Together they satisfy the Friedmann equation.

3. Where Do the Zeta Zeros Emerge From?

The Euler product representation of the zeta function:

$$\zeta(s) = \prod_p \frac{1}{(1 - p^{-s})} \quad \text{for } \operatorname{Re}(s) > 1$$

For $s = 1/2 + i*t$ (on the critical line):

$$\text{factor}_p(t) = \frac{1}{(1 - p^{-1/2} * e^{-i*t*\log(p)})}$$

Each prime p contributes an OSCILLATION at frequency:

$$f_p = \log(p) / (2*\pi)$$

The MODULUS of each factor oscillates between:

$$\text{MIN: } \frac{1}{(1 + p^{-1/2})} \quad (\text{constructive interference})$$

$$\text{MAX: } \frac{1}{(1 - p^{-1/2})} \quad (\text{destructive interference})$$

The PRODUCT converges when oscillations cancel on average.

The PRODUCT diverges when they reinforce.

The ZEROS are where the product = 0, i.e., perfect cancellation.

THE ZETA ZEROS EMERGE FROM:

The $0/0$ of the Euler product at $\operatorname{Re}(s) = 1/2$.

Each prime contributes a frequency $f_p = \log(p)/(2*\pi)$.

The zeros are the EIGENFREQUENCIES where these prime oscillations cancel EXACTLY.

The critical line $\operatorname{Re}(s) = 1/2$ is the BALANCE POINT between convergence and divergence.

4. Universe Growth = Zeta Zero Spectrum

The Friedmann equation with the prime spectrum:

$$H^2 = (8\pi G/3) * \rho_{\text{total}}$$

$$\rho_{\text{total}} = \rho_{\text{matter}} + \rho_{\text{radiation}} + \rho_{\text{Lambda}} + \rho_{\text{prime}}$$

$$\rho_{\text{prime}}(a) = \sum_n A_n * \cos(\gamma_n * \log(a))$$

Each zeta zero γ_n creates an OSCILLATION in the expansion rate as a function of $\log(a)$.

This is the origin of Baryon Acoustic Oscillations (BAO):

- The CMB power spectrum has peaks at specific multipoles
- These peaks correspond to the zeta zero frequencies
- The first peak is at $l \sim 220$, $\gamma_1 \sim 14.13$

THE UNIVERSE GROWS BY SUPERIMPOSING PRIME FREQUENCIES:

$$a(t) = a_0 * \exp(H_0 t) * \prod_n (1 + A_n * \cos(\gamma_n t))$$

Each zeta zero γ_n is a STANDING WAVE in the cosmic fluid.

The primes are the EIGENSTATES.

The zeros are the EIGENVALUES.

5. Explicit Formula = Friedmann Equation

The von Mangoldt explicit formula for primes:

$$\psi(x) = x - \sum_{\rho} x^{\rho}/\rho - \log(2\pi) - 1/2 \log(1-x^{-2})$$

Cosmological interpretation ($x = \text{volume} = a^3$):

$$\begin{aligned} \psi(a^3) &= a^3 && \leftarrow \text{MATTER (volume)} \\ &- \sum_n a^{3/2} * e^{i\gamma_n \log(a^3)} / \rho_n && \leftarrow \text{PRIME OSCILLATIONS (BAO)} \\ &- \log(2\pi) && \leftarrow \text{BOUNDARY (conformal)} \\ &- 1/2 \log(1-a^{-6}) && \leftarrow \text{CORRECTION (finite-size)} \end{aligned}$$

The Friedmann equation becomes:

$$\begin{aligned} H^2 &= (8\pi G/3) * d(\psi)/d(a^3) \\ &= (8\pi G/3) * [1 - \sum_n (3/2 * a^{-3/2} * e^{i\gamma_n \log(a^3)} / \rho_n)] \end{aligned}$$

The Hubble parameter H has OSCILLATIONS from the zeta zeros!

These are the BAO oscillations in the expansion rate.

6. The First 20 Cosmic Harmonics

The zeta zeros as frequencies, amplitudes, and phases:

n	γ_n	Frequency	Amplitude	Phase
1	14.1347	2.2496	10.4933	-1.560
2	21.0220	3.3458	7.0579	-1.813
3	25.0109	3.9806	5.9328	+0.372
4	30.4249	4.8423	4.8774	-2.040
5	32.9351	5.2418	4.5057	-2.072
6	37.5862	5.9820	3.9483	+0.455
7	40.9187	6.5124	3.6268	+2.363
8	43.3271	6.8957	3.4252	+1.313
9	48.0052	7.6403	3.0914	-2.172
10	49.7738	7.9218	2.9816	+2.948

These are the HARMONICS of the universe.

Each zero γ_n is a frequency at which the prime distribution oscillates. The universe grows by superimposing these oscillations.

7. The Complete 0/0 Universe

THE BEGINNING: $\log_0(0) = x$

$0^x = 0$ for all $x > 0$

The singularity at the origin of being.

The universe is the removable value.

THE END: $\log_1(1) = 0$

$1^0 = 1$

The tautology at the boundary of knowledge.

The universe collapses to the fixed point.

THE MIDDLE: The zeta zeros γ_n

The harmonics between beginning and end.

Each zero is a standing wave in the cosmic fluid.

The universe grows by superimposing them.

THE UNIVERSE IS THE REMOVABLE VALUE OF $\log_0(0)$.

THE ZETA ZEROS ARE THE FREQUENCIES OF THIS REMOVAL.

Every prime p contributes a frequency $f_p = \log(p)/(2\pi i)$.

The zeros are where these frequencies cancel perfectly.

The product of all prime oscillations is the UNIVERSE.

The Three 0/0 Levels

LEVEL 1: $\log_0(0) = x$ (the origin)

$0^x = 0$ for all $x > 0$

=> The universe exists because 0^x is the SAME for all x

=> The removable value is the initial condition

LEVEL 2: $\zeta(s)$ at the critical line (the interior)

$\zeta(1/2 + i\gamma_n) = 0$

=> The zeros are the 0/0 of the Euler product

=> Each zero is a harmonic of the prime spectrum

LEVEL 3: $\log_1(1) = 0$ (the boundary)

$1^0 = 1$

=> The tautology: the universe is what it is

=> The removable value is the FINAL condition

The universe begins at $\log_0(0)$ (the singularity).

It grows through the zeta zero harmonics (the interior).

It ends at $\log_1(1)$ (the tautology).

The entire history of the universe is contained
in the interval between these two 0/0 singularities.

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